

RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY
BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
MODEL EXAMINATION

Year/ Sem: I/I
Date: 10.02.2023

Max. Marks: 75
Max. Time: 3 Hours

PART I – (10 x 2 = 20 marks)

Answer ALL questions

SECTION – A
(ELECTRICAL)

1. Two resistors 6Ω and 6Ω are connected in parallel. If the total current is 3A, find the current through each resistor.
2. State Kirchoff's laws.
3. Define average value and RMS value.
4. Give the relation between line voltage and phase voltage for star and delta connection.
5. State Faraday's law.

SECTION – B
(ELECTRONICS)

6. Define Efficiency and ripple factor of rectifier?
7. Why are using common emitter configurations in all the circuits? Give reason?
8. Derive the expressions of Boolean Algebra
$$\overline{(AB + BC + CA)} = \bar{A} \bar{B} + \bar{B} \bar{C} + \bar{C} \bar{A}$$
9. What is ISDN? Give application and limitation?
10. Comparison of analog and digital signals?

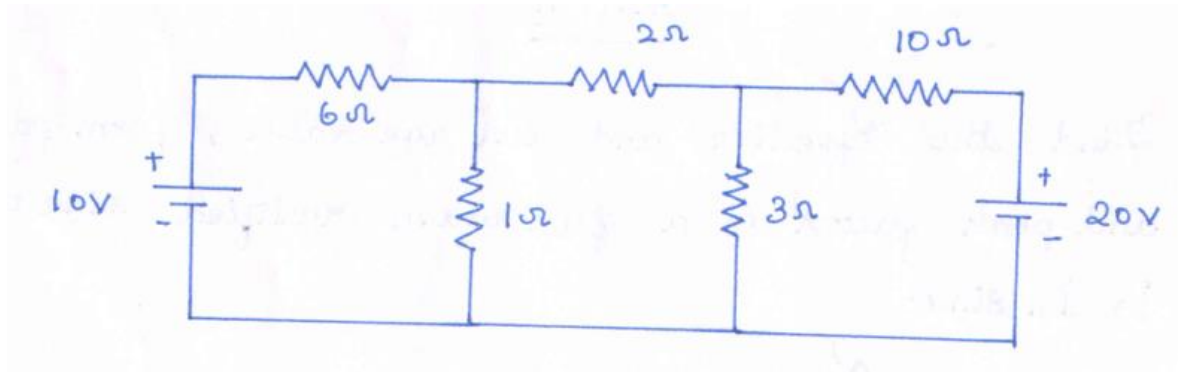
PART II – (55 marks)
SECTION – A
(ELECTRICAL) – (27 Marks)
Answer all the questions

UNIT - I

11. a) Derive the effective resistance for series circuit. (4)
b) Derive the expression for the delta to star transformation of three resistors. (5)

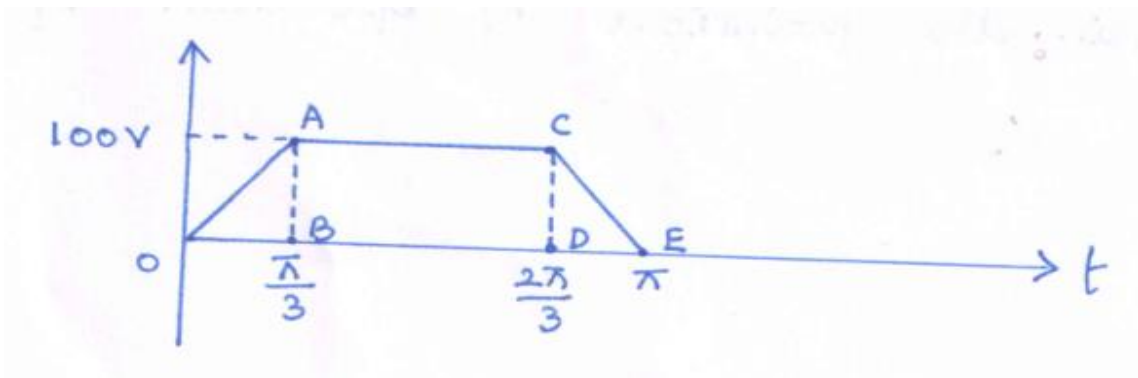
(OR)

12. Find the current through 2Ω resistor in the network using node voltage analysis. (9)



UNIT – II

13. Find the average value, effective value, form factor and peak factor for the given trapezoidal waveform. (9)



(OR)

14. Explain the two wattmeter method of measurement of three phase power with neat phasor diagram. (9)

UNIT – III

15. Discuss the construction and operation of transformer. (9)

(OR)

16. Explain with neat block diagram thermal power plant. (9)

SECTION – B
(ELECTRONICS) – (28 Marks)

Answer all the questions

UNIT - I

17. Draw the circuit diagram of a full wave rectifier network supplying a resistive load. Explain the operation of network with its analysis. (10)
- OR
18. Explain the working of BJT in CB configuration with suitable diagram and characteristics curves. (10)

UNIT - II

19. Draw the diagram of Full Subtractor and explain its operation. Write the truth table of Full Subtractor. (9)
- OR
20. Write detailed notes about the operation of shift register with necessary diagram. (9)

UNIT - III

21. Draw the block diagram of microwave communication system and explain its operation. (9)
- OR
22. Explain the block diagram of Integrated Services Digital Network. (9)